

## In-Process QC Checklist for Stationary Kavach Installation

QC Engineer Name: sushma

v1.2

| Station ID | Station Name | Zone | Division   | Section Name | Initial Date | Updated Date |
|------------|--------------|------|------------|--------------|--------------|--------------|
| 25285282   | Station      | ER   | Howrah-COO | Howrah-COO   | 23/03/2026   | 23/03/2026   |

|                     |                                    |                                                   |
|---------------------|------------------------------------|---------------------------------------------------|
| Installation Status | <input type="checkbox"/> Completed | <input checked="" type="checkbox"/> Not Completed |
|---------------------|------------------------------------|---------------------------------------------------|

| Total Points | Closed Points | Open Points |
|--------------|---------------|-------------|
| 154          | 153           | 1           |

## 1.0 Serial Number Verification

| S.No | Units                        | Status   | Observation | Image |
|------|------------------------------|----------|-------------|-------|
| 1.1  | Stationary Kavach Unit       | Matching |             |       |
| 1.2  | Peripheral Processing Card 1 | Matching |             |       |
| 1.3  | Peripheral Processing Card 2 | Matching |             |       |
| 1.4  | Vital Computer Card -1       | Matching |             |       |
| 1.5  | Vital Computer Card -2       | Matching |             |       |
| 1.6  | Vital Computer Card -3       | Matching |             |       |
| 1.7  | Voter Card -1                | Matching |             |       |
| 1.8  | Voter Card -2                | Matching |             |       |
| 1.9  | Vital Gateway Card 1         | Matching |             |       |
| 1.10 | Vital Gateway Card 2         | Matching |             |       |
| 1.11 | Vital Gateway Card 3 (NMS)   | Matching |             |       |
| 1.12 | EI Gateway-1                 | Matching |             |       |
| 1.13 | EI Gateway-2                 | Matching |             |       |
| 1.14 | Field Scanner Card 1         | Matching |             |       |
| 1.15 | Field Scanner Card 2         | Matching |             |       |
| 1.16 | Field Scanner Card 3         | Matching |             |       |
| 1.17 | Field Scanner Card 4         | Matching |             |       |
| 1.18 | Field Scanner Card 5         | Matching |             |       |
| 1.19 | Field Scanner Card 6         | Matching |             |       |
| 1.20 | Field Scanner Card 7         | Matching |             |       |

| S.No | Units                             | Status   | Observation | Image |
|------|-----------------------------------|----------|-------------|-------|
| 1.21 | Field Scanner Card 8              | Matching |             |       |
| 1.22 | RIU communication card 1-Host     | Matching |             |       |
| 1.23 | RIU communication card 2-Host     | Matching |             |       |
| 1.24 | RS 232-OFC converter 1 (STATION)  | Matching |             |       |
| 1.25 | RS 232-OFC converter 2 (STATION)  | Matching |             |       |
| 1.26 | RS 485-OFC converter (STATION)    | Matching |             |       |
| 1.27 | FIU Termination Card 1            | Matching |             |       |
| 1.28 | FIU Termination Card 2            | Matching |             |       |
| 1.29 | FIU Termination Card 3            | Matching |             |       |
| 1.30 | FIU Termination Card 4            | Matching |             |       |
| 1.31 | FIU Termination Card 5            | Matching |             |       |
| 1.32 | FIU Termination Card 6            | Matching |             |       |
| 1.33 | FIU Termination Card 7            | Matching |             |       |
| 1.34 | FIU Termination Card 8            | Matching |             |       |
| 1.35 | DPS Card 1                        | Matching |             |       |
| 1.36 | DPS Card 2                        | Matching |             |       |
| 1.37 | GPS & GSM Antenna-1               | Matching |             |       |
| 1.38 | GPS & GSM Antenna-2               | Matching |             |       |
| 1.39 | SMOCIP Unit                       | Matching |             |       |
| 1.40 | SMOCIP Termination Panel          | Matching |             |       |
| 1.41 | Station Termination Panel         | Matching |             |       |
| 1.42 | Stationary PDU Box                | Matching |             |       |
| 1.43 | IPS PDU Box                       | Matching |             |       |
| 1.44 | DC-DC Converter (Relay racks)     | Matching |             |       |
| 1.45 | RTU-1                             | Matching |             |       |
| 1.46 | RTU-2                             | Matching |             |       |
| 1.47 | Station Radio Power Supply card-1 | Matching |             |       |
| 1.48 | Station Radio Power Supply card-2 | Matching |             |       |
| 1.49 | Next Gen/. Cal Amp Radio Modem-1  | Matching |             |       |
| 1.50 | Next Gen/. Cal Amp Radio Modem-2  | Matching |             |       |
| 1.51 | RS 232-OFC converter 1 (RTU-1)    | Matching |             |       |
| 1.52 | RS 232-OFC converter 2 (RTU-2)    | Matching |             |       |

| S.No | Units                           | Status   | Observation | Image |
|------|---------------------------------|----------|-------------|-------|
| 1.53 | RS 485-OFC converter (SM-OCIP)  | Matching |             |       |
| 1.54 | RIU                             | Matching |             |       |
| 1.55 | RIU Power Supply Card-1         | Matching |             |       |
| 1.56 | RIU Power Supply Card-2         | Matching |             |       |
| 1.57 | RIU communication card 1-Remote | Matching |             |       |
| 1.58 | RIU communication card 2-Remote | Matching |             |       |
| 1.59 | Field Scanner Card 1            | Matching |             |       |
| 1.60 | Field Scanner Card 2            | Matching |             |       |
| 1.61 | Field Scanner Card 3            | Matching |             |       |
| 1.62 | Field Scanner Card 4            | Matching |             |       |
| 1.63 | RIU Battery Charge Cum Filter-1 | Matching |             |       |
| 1.64 | RIU Battery Charge Cum Filter-2 | Matching |             |       |

## 2.0 Building

| S.No | Aspect                | Requirement                                                                                                                                         | Status    | Observation | Image |
|------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------|
| 2.1  | KAVACH Building Works | Building Commissioning Report from the contractor shall be available with the customer's sign-off, and there shall be no open points in the report. | Available |             |       |

## 3.0 Tower

| S.No | Aspect             | Requirement                                                                                                                                      | Status    | Observation | Image |
|------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------|
| 3.1  | Tower Installation | Tower Commissioning Report from the contractor shall be available with the customer's sign-off, and there shall be no open points in the report. | Available |             |       |

## 4.0 Power Supply

| S.No                                    | Aspect        | Requirement                                                                                                                                                                                                   | Status | Observation | Image |
|-----------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| <b>4.1 PDU(Power Distribution Unit)</b> |               |                                                                                                                                                                                                               |        |             |       |
| 4.1                                     | Installation  | IPS PDU unit shall be mounted on the wall using M8 insulators, secured with M8 bolts, and tightened to a torque of 20 Nm as per diagram 5 16 76 0053                                                          | Yes    |             |       |
| 4.1.1                                   | Installation  | Station PDU unit shall be mounted on the wall using M8 insulators, secured with M8 bolts, and tightened to a torque of 20 Nm as per diagram 5 16 76 0054                                                      | Yes    |             |       |
| 4.1.2                                   | Cabling       | All external cables entering the PDU shall pass through cable glands and ensure no cable entry opening shall be used without a cable gland.                                                                   | Yes    |             |       |
| 4.1.3                                   | Cabling       | Output connections shall be maintained as per the Station PDU schematic diagram 5 16 49 0671                                                                                                                  | Yes    |             |       |
| 4.1.4                                   | Cabling       | Ensure lugs with sleeves / Ferrules are properly crimped and inserted into the terminal; no loose strands shall be left.                                                                                      | Yes    |             |       |
| 4.1.5                                   | Earthing      | PDU units shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.                              | Yes    |             |       |
| 4.1.6                                   | Earthing      | Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.                                                                                                 | Yes    |             |       |
| 4.1.7                                   | Functionality | Functional testing shall be performed as per the PDU test procedure 5 53 69 0001.                                                                                                                             | Yes    |             |       |
| <b>4.2 DC-DC Converter</b>              |               |                                                                                                                                                                                                               |        |             |       |
| 4.2                                     | Installation  | DC-DC Converter unit shall be installed as per the approved floor plan drawing, mounted on floor using M10 insulators, secured with M10 bolts, and tightened to a torque of 40 Nm as per diagram 5 16 76 0055 | Yes    |             |       |

| S.No  | Aspect        | Requirement                                                                                                                                                                 | Status | Observation | Image |
|-------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 4.2.1 | Cabling       | All external cables entering the unit shall pass through cable glands and ensure no cable entry opening shall be used without a cable gland.                                | Yes    |             |       |
| 4.2.2 | Earthing      | Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied. | Yes    |             |       |
| 4.2.3 | Earthing      | Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.                                                               | Yes    |             |       |
| 4.2.4 | Functionality | DC-DC converter output voltage shall be minimum 24 V DC, +/- 5% (22.8 VDC to 25.2 VDC)                                                                                      | Yes    |             |       |

## 5.0 Kavach Equipment

| S.No            | Aspect       | Requirement                                                                                                                                                                                               | Status | Observation | Image |
|-----------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 5.1 Kavach Unit |              |                                                                                                                                                                                                           |        |             |       |
| 5.1             | Installation | Kavach unit shall be installed as per the approved floor plan drawing, mounted on the floor using M10 insulators, secured with M10 bolts, and tightened to a torque of 40 Nm as per diagram 5 16 76 0056. | Ok     |             |       |
| 5.1.1           | Cabling      | All external cables shall enter through cable glands only. No unused cable entries left open. Ensure mill connector shall be locked properly.                                                             | Ok     |             |       |
| 5.1.2           | Earthing     | Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.                               | Ok     |             |       |
| 5.1.3           | Earthing     | Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.                                                                                             | Ok     |             |       |
| 5.1.4           | Earthing     | Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.                                                                                             | Ok     |             |       |

| S.No                                                               | Aspect           | Requirement                                                                                                                                                                                                                 | Status | Observation | Image |
|--------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 5.1.5                                                              | Termination Unit | Station Kavach Termination Unit shall be wall-mounted near the Kavach unit using insulators, secured with M8 bolts, and tightened to a torque of 20 Nm as per diagram 5 16 76 0045.                                         | Ok     |             |       |
| 5.1.6                                                              | Termination Unit | OFC cables for SMOCIP and RTU shall be spliced as per diagram 5 16 49 0559, and proper bunching and routing shall be ensured.                                                                                               | Ok     |             |       |
| <b>5.2 SMOCIP(Station Master's Operation-Cum-Indication Panel)</b> |                  |                                                                                                                                                                                                                             |        |             |       |
| 5.2                                                                | Installation     | SMOCIP shall be installed in the Station Master's room at an ergonomic height. The panel shall be securely fixed using M6 screws and tightened to a torque of 8 Nm with torque marking applied, as per diagram 5 16 76 0040 | Ok     |             |       |
| 5.2.1                                                              | Termination Unit | SMOCIP Termination Unit shall be wall-mounted near the SMOCIP unit, using insulators, secured with M6 bolts, and tightened to a torque of 8 Nm as per diagram 5 16 76 0046                                                  | Ok     |             |       |
| 5.2.2                                                              | Termination Unit | Power and OFC cables from Kavach termination unit shall be terminated as per diagram 5 16 49 0559.                                                                                                                          | Ok     |             |       |
| 5.2.3                                                              | Termination Unit | OFC cables of SMOCIP shall be spliced as per diagram 5 16 49 0559, and proper bunching and routing shall be ensured.                                                                                                        | Ok     |             |       |
| 5.2.4                                                              | Earthing         | Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.                                                 | Ok     |             |       |
| 5.2.5                                                              | Earthing         | Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.                                                                                                               | Ok     |             |       |
| 5.2.6                                                              | Functionality    | System Health LED shall blink and ensure SYSTEM OK along with the respective station name shall be displayed on the Display.                                                                                                | Ok     |             |       |

| S.No                                  | Aspect        | Requirement                                                                                                                                                                                                                                      | Status         | Observation | Image |
|---------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------|-------|
| 5.2.7                                 | Functionality | Verify that pressing the SOS and Common buttons on the SM-OCIP increments the mechanical counter by one.                                                                                                                                         | Ok             |             |       |
| 5.2.8                                 | Checksum      | Verify the checksums as per the FAT certificate.                                                                                                                                                                                                 | Matching       |             |       |
| <b>5.3 GPS/GSM Antennas</b>           |               |                                                                                                                                                                                                                                                  |                |             |       |
| 5.3                                   | Installation  | Two antennas shall be installed on the Kavach room rooftop with a minimum separation of 3 meters, grouting shall be carried out as per diagram 5 16 67 0039, and torque of 10 Nm shall be applied for M6 fasteners with torque marking provided. | Ok             |             |       |
| 5.3.1                                 | Installation  | No obstruction above antennas like tree branches, sun-shades, and ensure open to sky etc.                                                                                                                                                        | Ok             |             |       |
| 5.3.2                                 | Cabling       | Antenna cables shall be routed via diverse paths.                                                                                                                                                                                                | Not Applicable |             |       |
| 5.3.3                                 | Cabling       | Separate conduits shall be used and Roof conduits shall be sealed against dust, water, and insects.                                                                                                                                              | Ok             |             |       |
| 5.3.4                                 | Cabling       | In each antenna, the GPS and GSM cables shall be connected to their respective connectors as per the labels provided on the antenna.                                                                                                             | Ok             |             |       |
| <b>5.4 RIU(Remote Interface Unit)</b> |               |                                                                                                                                                                                                                                                  |                |             |       |
| 5.4                                   | Installation  | RIU shall be installed on floor using M10 insulators, secured with M10 bolts, and tightened to a torque of 40 Nm as per diagram 5 16 76 0057.                                                                                                    | Ok             |             |       |
| 5.4.1                                 | Cabling       | All external cables entering into the RIU unit shall pass through cable glands and ensure no cable entry opening shall be used without a cable gland.                                                                                            | Not Applicable |             |       |
| 5.4.2                                 | Cabling       | OFC patch cords shall be properly tagged to identify default and standby links.                                                                                                                                                                  | Ok             |             |       |

| S.No  | Aspect                | Requirement                                                                                                                                                                 | Status | Observation | Image |
|-------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 5.4.3 | Earthing              | Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied. | Ok     |             |       |
| 5.4.4 | Earthing              | Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.                                                               | Ok     |             |       |
| 5.4.5 | FDMS Box installation | FDMS Box shall be installed in the 15U/17U rack of the RIU with proper wall mounting, and OFC splicing shall be carried out as per the network drawing.                     | Ok     |             |       |

## 6.0 RF Communication Equipment On Tower

| S.No                             | Aspect                | Requirement                                                                                                                                                                        | Status | Observation | Image |
|----------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| <b>6.1 RTU(Radio Tower Unit)</b> |                       |                                                                                                                                                                                    |        |             |       |
| 6.1                              | RTU Fixing            | Both RTUs shall be firmly secured to the tower platform using M12 bolts and nuts, and a torque of 85 Nm shall be applied as per diagram 5 16 67 0983.                              | Ok     |             |       |
| 6.1.1                            | RTU Fixing            | Ensure RTU doors are fully closed and locked.                                                                                                                                      | Ok     |             |       |
| 6.1.2                            | RTU Earthing          | RTU shall be properly earthed by connecting a 35 sq.mm green to GI strip earthing conductor from the RTU earthing bolt to the designated earth pit-4, as per diagram 5 16 76 0043. | Ok     |             |       |
| 6.1.3                            | RTU Earthing          | Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.                                                                      | Ok     |             |       |
| 6.1.4                            | OFC cable termination | OFC cable from the Relay Room shall be spliced and terminated in the splice holder inside the RTU.<br>(Ref. Drawing: 5 16 49 0559)                                                 | Ok     |             |       |
| 6.1.5                            | OFC cable termination | OFC cables for RTU shall be spliced as per diagram 5 16 49 0559 and ensure bunching and routing shall be done properly.                                                            | Ok     |             |       |

| S.No                               | Aspect                            | Requirement                                                                                                                                             | Status         | Observation | Image |
|------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------|-------|
| 6.1.6                              | 110V Power cable termination      | Cable glands used for 110 V DC power cable entry into RTU shall be firmly tightened.                                                                    | Ok             |             |       |
| 6.1.7                              | 110V Power cable termination      | 110 V DC power cables shall be terminated inside RTU as per approved drawing 5 16 49 0672.                                                              | Ok             |             |       |
| 6.1.8                              | 110V Power cable termination      | Ensure lugs with sleeves / Ferrules are properly crimped and inserted into the terminal; no loose strands shall be left.                                | Ok             |             |       |
| 6.1.9                              | Cabling                           | LMR 600 connection with proper routing; and clamping; No Joints shall be done, as per Tower SOP 5 16 90 0018.                                           | Ok             |             |       |
| <b>6.2 RF Antenna Installation</b> |                                   |                                                                                                                                                         |                |             |       |
| 6.2                                | RF antenna installation and Audit | RF antenna installation and orientation shall be done as per 10.2dBi omni-directional antenna diagram 5 16 67 0983.                                     | Ok             |             |       |
| 6.2.1                              | RF antenna installation and Audit | RF antenna installation audit report from the installation contractor shall be available in WFMS and there shall be no open points in the audit report. | Not Applicable |             |       |

## 7.0 OFC Networking Rack

| S.No  | Aspect       | Requirement                                                                                                               | Status         | Observation | Image |
|-------|--------------|---------------------------------------------------------------------------------------------------------------------------|----------------|-------------|-------|
| 7.1   | Installation | Network rack shall be installed as per approved floor plan and installation shall be done as per diagram 5 16 76 0058.    | Ok             |             |       |
| 7.1.1 | Installation | Patch cord routing shall be neat & bend radius to be maintained.                                                          | Not Applicable |             |       |
| 7.2   | Cabling      | All external cables entering the network rack shall pass through the grommets.                                            | Ok             |             |       |
| 7.2.1 | Cabling      | OFC cables shall be marked using naming tie-tags for easy identification of default and standby links.                    | Ok             |             |       |
| 7.3   | Earthing     | All networking modules inside the rack shall be connected to the rack chassis using 2.5 sq.mm green/yellow earthing wire. | Ok             |             |       |

| S.No  | Aspect                                     | Requirement                                                                                           | Status | Observation | Image |
|-------|--------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 7.3.1 | Earthing                                   | Network rack shall be connected to ring earth using a 10 sq.mm green/yellow earthing wire.            | Ok     |             |       |
| 7.3.2 | Earthing                                   | Bolts shall be tightened to a torque of 8 Nm, and torque marking shall be applied using yellow paint. | Ok     |             |       |
| 7.4   | FDMS Box Installation in Network Rack      | OFC cables shall be spliced in the FDMS box as per network drawing.                                   | Ok     |             |       |
| 7.4.1 | FDMS Box Installation in Network Rack      | FDMS boxes shall be clearly marked to identify up and down links.                                     | Ok     |             |       |
| 7.5   | OFC cable continuity check, after splicing | OTDR test reports shall be available.                                                                 | Ok     |             |       |

## 8.0 Relay Rack

| S.No  | Aspect                      | Requirement                                                                                                                                                                                   | Status         | Observation | Image |
|-------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------|-------|
| 8.1   | Installation                | Relay rack shall be installed as per the approved Floor Plan drawing, mounted on insulators, and secured to the floor by grouting as per diagram 5 16 76 0059.                                | Ok             |             |       |
| 8.2   | Wiring                      | Labeling sleeve shall be used to identify wiring with rack number, row number, relay number & contact type, Labelling shall be provided to relay contact wires at FTC PCBA of Station Kavach. | Not Applicable |             |       |
| 8.2.1 | Wiring                      | For EI Stations, verify all connections as per the approved EI Interface diagrams (vendor specific).                                                                                          | Ok             |             |       |
| 8.2.2 | Wiring                      | WAGO terminal details shall be as per interface circuit diagram 5 16 49 0685.                                                                                                                 | Ok             |             |       |
| 8.3   | Continuity Test / Bell Test | Completed Station Analyser and Bell Test reports shall be available.                                                                                                                          | Ok             |             |       |
| 8.4   | Earthing                    | Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.                   | Ok             |             |       |

| S.No  | Aspect   | Requirement                                                                                                   | Status | Observation | Image |
|-------|----------|---------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 8.4.1 | Earthing | Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied. | Ok     |             |       |

## 9.0 Earthing

| S.No | Aspect                       | Requirement                                                                                                                                                           | Status    | Observation | Image |
|------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------|
| 9.1  | Installation                 | Earthing shall be done as per RDSO Spec RDSO/SPN/197/2008 and earth pit placement shall be done as per diagram 5 16 76 0043 .                                         | Ok        |             |       |
| 9.2  | Earth Resistance Measurement | Earth resistance shall be measured using a calibrated earth resistance tester. The measured value shall be less than 1 Ohm for Stationary Kavach equipment earth pit. | Ok        |             |       |
| 9.3  | Earth Resistance Measurement | Check the value of earth resistance at earth of the radio tower.                                                                                                      | Ok        |             |       |
| 9.4  | Test Reports                 | Earth resistance test reports shall be available.                                                                                                                     | Available |             |       |
| 9.5  | Labeling                     | All earth pits, earthing conductors, and earth points shall be clearly labelled and identifiable.                                                                     | Ok        |             |       |

## 10.0 Indoor Wiring / Cabling

| S.No | Aspect            | Requirement                                                                                                                    | Status | Observation | Image |
|------|-------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 10.1 | Redundant cabling | Cabling shall be done as per station connectivity diagram 5 16 49 0614 Segregation of power & communication cables.            | Ok     |             |       |
| 10.2 | SMOCIP            | 12 Core Signaling cable shall be used for button, counter & power supply. OFC armoured cables shall be used for communication. | Ok     |             |       |
| 10.3 | No Joints         | No joints permitted except inside junction boxes or panels.                                                                    | Ok     |             |       |
| 10.4 | Color Coding      | Colour codes shall be followed for phase, neutral, and earth.                                                                  | Ok     |             |       |

| S.No | Aspect                   | Requirement                                                           | Status | Observation | Image |
|------|--------------------------|-----------------------------------------------------------------------|--------|-------------|-------|
| 10.5 | Tagging/Labeling         | Cables shall be tagged at both ends.                                  | Ok     |             |       |
| 10.6 | Termination & Connection | Proper lugs/ferrules shall be used and terminals tightened correctly. | Ok     |             |       |

## 11.0 Outdoor Cabling

| S.No | Aspect                                                       | Requirement                                                                             | Status | Observation | Image |
|------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------|-------------|-------|
| 11.1 | Cable route plan for OFC & Power cable (Relay Room to Tower) | Railway-approved outdoor signalling cable route plan shall be available.                | Ok     |             |       |
| 11.2 | Cable route markers                                          | Route markers shall be installed and clearly visible.                                   | Ok     |             |       |
| 11.3 | Cable route markers                                          | Where contractually required, underground RFID markers shall be installed and verified. | Ok     |             |       |

## 12.0 RFID Tags

| S.No | Aspect                                              | Requirement                                                                                                                                                                                                  | Status    | Observation | Image |
|------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------|
| 12.1 | RFID Tag Installation                               | Ensure the main RFID tag is installed in the direction specified in the RFID tag layout, and a duplicate RFID tag shall be installed at a distance of 3–5 meters from the main tag, except for turnout tags. | Ok        |             |       |
| 12.2 | RFID Tag Installation                               | RFID tag mounting shall be firm and secure, ensuring no gap or mechanical play as per RFID tag installation procedure 5 53 76 0055.                                                                          | Ok        |             |       |
| 12.3 | RFID Tag data Validation and Placement verification | System-generated RFID data validation report and Placement verification report shall be available.                                                                                                           | Available |             |       |

## 13.0 Components Inspection

| S.No | Aspect                                                              | Requirement                                                                                                                                       | Status | Observation | Image |
|------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 13.1 | Materials Inspection (Which are not Inspected by RDSO or Consignee) | Verify that all subcontractors and HBL personnel use only approved make and part numbers as per the List of Acceptable I&C Materials EG-EN-FT-34. | Ok     |             |       |

**Note : This QC report is generated by the system and does not need a physical signature.**